

Ziying (Emma) Li

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EDUCATION

Zhejiang University

Master of Engineering in Software Engineering

Advisor & Collaborator: Xiaowei Zhou, Xuequan Lu, Xinkui Zhao

Hangzhou, Zhejiang

September 2024 – June 2027 (Expected)

University of Illinois Urbana-Champaign

Bachelor of Science in Computer Engineering

Urbana, IL

September 2020 – May 2024

Zhejiang University

Bachelor of Engineering in Electronic and Computer Engineering

Hangzhou, Zhejiang

September 2020 – May 2024

HIGHLIGHTS

- **Theory-to-system research on stochastic transport for generation:** first-author [NeurIPS 2025](#) paper formulating text-to-3D as explicit Schrödinger-Bridge transport from rendering to target, in place of the noise-anchored bridge implicit in prior objectives, sustaining state-of-the-art fidelity at guidance scales where they degrade; an ICLR 2027 submission supervising the conditional velocity covariance the standard objective ignores, with the flow-matching optimum exactly preserved.
- **Hands-on experience with scalable DiT-style flow models:** Co-developed [Hi3D 3.0](#), a 6.6B-parameter image-to-3D foundation model built on 3D VAE with an MMDiT-style flow-matching backbone, with focus on structured 3D latents, semantic conditioning, and trajectory-level objectives.

SELECTED PUBLICATIONS

Walking the Schrödinger Bridge: A Direct Trajectory for Text-to-3D Generation.

Ziying Li, Xuequan Lu, Xinkui Zhao, Guanjie Cheng, Shuiguang Deng, and Jianwei Yin

[NeurIPS 2025, Poster](#) [pdf] [github] [Video] [Slide]

Keywords: Diffusion models, Schrödinger Bridge, text-to-3D generation, 3D Gaussian Splatting.

Hi3D 3.0 Technical Report: Next Generation 3D-Native Foundation Model for Image-to-Geometry Generation.

Ziying Li, Yiyang Chen, Shengchu Zhao, Huiang He, Jiaqi Wu, Yibo Luo, Hu Zhang, Heliang Zheng, Rongfei Jia

[Technical Report, scheduled for arXiv release in Sep. 2026.](#) [Project Page] [Video] [Website]

Keywords: 3D generative modeling, flow matching, 3D VAE, image-to-3D generation, representation learning.

RESEARCH EXPERIENCE

Research Intern

Math Magic

A leading Generative AI startup pioneering state-of-the-art 3D foundation models (Hi3D)

Dec. 2025 – Present

Advisor: Xiaowei Zhou; Project Lead: Heliang Zheng

- Co-developed Hi3D 3.0, a 6.6B-parameter native image-to-3D foundation model: designed its two-stage vecset generative space and co-designed the semantic conditioning and representation alignment objectives of its MMDiT flow-matching backbone.
- Designed the two-stage generative space so that sparse structure is classified, never generated: a TRELIS-style sparse convolutional autoencoder compresses 1024^3 FlexiCubes fields into 128^3 sparse latents at roughly $2000\times$ compression; a vecset VAE rebuilt from Hunyuan3D-2.1 tokenizes the variable-length field into a fixed 4096×64 KL-regularized set; diffusion models only fixed-size tensors.
- Unified supervision across both compression stages, structure as classification and continuous geometry as regression: the sparse autoencoder recovers occupancy coarse to fine and confines rendering losses to the deformation branch, sidestepping the topology flips that destabilize end-to-end isosurface training; the vecset VAE anchors each latent token to a farthest-point-sampled 3D coordinate and decodes through dual heads, classifying voxel occupancy and regressing per-voxel features.
- Built the flow model's REPA teacher instead of aligning across the 2D-to-3D gap: canonical coordinate maps assign each multi-view DINOv3 patch a 3D position, and an axis-decomposed 3D RoPE cross-attention fuses the features into the VAE latent at those locations; a pre-KL centered-kernel-alignment loss then matches the latent's similarity structure to DINO, a criterion indifferent to feature dimension; the frozen aligned latent supervises denoiser features entirely within the 3D latent space.

- Designed the MMDiT image conditioning interface by matching mechanism to information type: dense DINOv3 patch tokens carry spatial correspondence through joint attention; pooled SigLIP2 embeddings steer global semantics through AdaLN; a modality-decoupled 3D RoPE rotates only 3D-to-3D query-key pairs.
- Attributed a class of generation failures to flow-matching transport dynamics rather than model capacity: averaging over conflicting branch velocities cancels modulus in the conditional mean field, producing a systematic velocity deficit; applied and validated MAFM, a modulus-matching auxiliary objective, and derived its exact compensation, proving its fixed point restores precisely the direction-dispersion component of conditional velocity variance while leaving length dispersion untouched.
- *Technical report will be released on arxiv in September.*

Research Assistant

Zhejiang University, Math Magic

May 2026 – Present

Advisor: Xiaowei Zhou

- Identified occluded-region failure in image-to-3D flow matching as blunted anisotropy of the model-implied conditional velocity covariance: the velocity Jacobian encodes this covariance through a second-order Tweedie identity, yet the standard objective leaves it entirely unsupervised.
- Proposed a directional second-moment matching loss that supervises the Jacobian-implied covariance against single-sample even-order targets; sign-free, aggregation-free, and head-free by construction, with the flow-matching optimum exactly preserved, so branch commitment is restored by sampling dynamics rather than injected as bias.
- Distilled three design axioms from systematic autopsies of failed second-moment objectives, even-order targets, conditional regression form, and a derivative-slot carrier, together with a zero-cost validation ladder of stationary-point audits, implied-versus-behavioral anisotropy headroom tests, and gradient-geometry tripwires.
- *Prepared to submit to ICLR 2027.*

Research Assistant

Zhejiang University, The University of Western Australia

Dec. 2024 – Jun. 2025

Advisor: Xuequan Lu, Xinkui Zhao

- Recast Score Distillation Sampling (SDS) as a degenerate Schrödinger Bridge: under a Gaussian terminal prior and a trivial forward Schrödinger factor, the bridge reverse score reduces to the pretrained diffusion score, identifying explicit endpoint control as the structure SDS forfeits.
- Proposed TraCe, a distillation framework that rebuilds the bridge between the two distributions the optimization actually connects: the current rendering occupies the noisy end, its text-conditioned one-step denoising serves as the target, and both endpoints refresh at every optimization step so the trajectory tracks the evolving 3D state.
- Derived and implemented the TraCe optimization objective by sampling intermediate states from the trackable between the current 3D rendering and denoised target, and trained a LoRA-adapted diffusion model under an annealed time schedule to learn the trajectory's score dynamics.
- Validated on 3D Gaussian Splatting across 83 prompts and 120 views, outperforming SDS, VSD, CSD, SDI, and ISM on CLIP Score, GPTEval3D, and ImageReward while maintaining high-fidelity generation under moderate CFG values.

SELECTED AWARDS AND HONORS

Member of the American Engineering Honor Society Tau Beta Pi	2023
Tau Beta Pi Outstanding Junior Award	2023
First Prize in the International Division of the Mathematical Contest in Modeling (MCM) (Top 7%)	2021
UIUC Dean's List	2021

SERVICE

Reviewer, NeurIPS 2026

Reviewer, IEEE Transactions on Multimedia

TECHNICAL SKILLS

Programming languages: Python, C/C++, CUDA, Bash

Tools & Training Frameworks: Pytorch, DeepSpeed, CUDA, DDP/NCCL, gradient checkpointing, Git, Docker

Generative Modeling: Diffusion/flow matching, Schrödinger Bridge, Optimal Transport, DiT/MMDiT, VAE, LoRA, Representation Alignment